

Quiz 9

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (3 points) Evaluate the scalar line integral $\int_C x \sin y \, ds$ where C is the line segment from $(0, 3)$ to $(4, 6)$.

$$\mathbf{X}(t) = (0, 3) + t((4, 6) - (0, 3)) = (0, 3) + t(4, 3)$$

$$= (4t, 3t+3), \quad 0 \leq t \leq 1$$

$$\Rightarrow \mathbf{X}'(t) = (4, 3) \quad \text{and} \quad \|\mathbf{X}'(t)\| = \sqrt{4^2 + 3^2} = 5$$

$$\begin{aligned} \int_C x \sin y \, ds &= \int_0^1 (4t) \sin(3t+3) (5 \, dt) = \int_0^1 20t \sin(3t+3) \, dt \\ &= \left[20t \cdot \left(-\frac{1}{3} \cos(3t+3)\right) \right]_{t=0}^{t=1} - \int_0^1 20 \cdot \left(-\frac{1}{3} \cos(3t+3)\right) \, dt \\ &= \left[-\frac{20}{3} t \cos(3t+3) \right]_{t=0}^{t=1} + \int_0^1 \frac{20}{3} \cos(3t+3) \, dt \\ &= -\frac{20}{3} \cos 6 + \left[\frac{20}{9} \sin(3t+3) \right]_{t=0}^{t=1} = -\frac{20}{3} \cos 6 + \frac{20}{9} (\sin 6 - \sin 3) \end{aligned}$$

Problem 2. (3 points) Evaluate the vector line integral $\int_{\mathbf{x}} \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{F}(x, y, z) = (x + y)\mathbf{i} + (y - z)\mathbf{j} + z^2\mathbf{k}$ and the path $\mathbf{x}(t) = t^2\mathbf{i} + t^3\mathbf{j} + t^2\mathbf{k}$, $0 \leq t \leq 1$.

$$\mathbf{X}'(t) = (2t, 3t^2, 2t)$$

$$\begin{aligned} \int_{\mathbf{x}} \mathbf{F} \cdot d\mathbf{s} &= \int_0^1 (t^2 + t^3, t^3 - t^2, t^4) \cdot (2t, 3t^2, 2t) \, dt \\ &= \int_0^1 (t^2 + t^3) \cdot (2t) + (t^3 - t^2) \cdot (3t^2) + t^4 \cdot (2t) \, dt \\ &= \int_0^1 (2t^3 + 2t^4 + 3t^5 - 3t^4 + 2t^5) \, dt \\ &= \int_0^1 (5t^5 - t^4 + 2t^3) \, dt \\ &= \left[\frac{5}{6} t^6 - \frac{1}{5} t^5 + \frac{1}{2} t^4 \right]_{t=0}^{t=1} \\ &= \frac{5}{6} - \frac{1}{5} + \frac{1}{2} = \frac{17}{15} \end{aligned}$$

Problem 3. (4 points) Use **Green's theorem** to find the area of the region D enclosed by the hypocycloid

$$\mathbf{x}(t) = (2\cos^3 t, 2\sin^3 t), \quad 0 \leq t \leq 2\pi.$$

$$\begin{aligned} \text{Area}(D) &= \frac{1}{2} \oint_D -y dx + x dy \quad \xrightarrow{\mathbf{x}'(t) = (-6\cos^2 t \sin t, 6\sin^2 t \cos t)} \\ &= \frac{1}{2} \int_0^{2\pi} (-2\sin^3 t)(-6\cos^2 t \sin t) dt \\ &\quad + (2\cos^3 t)(6\sin^2 t \cos t) dt \\ &= \frac{1}{2} \int_0^{2\pi} 12\sin^4 t \cos^2 t + 12\sin^2 t \cos^4 t dt \\ &= 6 \int_0^{2\pi} \sin^2 t \cos^2 t (\sin^2 t + \cos^2 t) dt \\ &= 6 \int_0^{2\pi} \sin^2 t \cos^2 t dt \\ &= \frac{3}{2} \int_0^{2\pi} \sin^2 2t dt \\ &= \frac{3}{2} \int_0^{2\pi} \frac{1 - \cos 4t}{2} dt \\ &= \frac{3}{2} \left[\frac{t}{2} - \frac{1}{8} \sin 4t \right]_{t=0}^{t=2\pi} \\ &= \frac{3}{2} \cdot 2\pi \cdot \frac{1}{2} = \frac{3}{2}\pi \end{aligned}$$